COURSE SLIDES · NON-WSQ

Sustainability Masterclass

Course Code: C752

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome and Outcomes

C752 · 2 days · 15 instructional hours

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About the Trainer



Your Trainer

To be completed by the trainer

NAME

DESIGNATION

QUALIFICATIONS

INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

FOCUS
Applied AI, sustainability data and business transformation

ROLE
Course facilitation, activity coaching and learner support

APPROACH
Evidence-led, practical and decision-focused

ORGANISATION
Tertiary Infotech Academy Pte Ltd

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INTRODUCTIONS

Let’s Know Each Other

1

You

Name, organisation and role

2

Experience

Exposure to ESG, operations, energy, finance or reporting

3

Challenge

One footprint or sustainability decision you want to improve

4

Outcome

What a useful carbon-reduction plan would enable

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HOUSEKEEPING

Ground Rules

1. Participate actively and ask questions.
2. Respect different operational perspectives.
3. Use fictionalised or approved business data.
4. Return punctually from breaks.
5. Keep phones silent outside learning tasks.
6. Challenge assumptions with evidence.

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LEARNING RESOURCES

Access Your Course Materials

1

LMS

Sign in at lms-tms.tertiaryinfotech.com

2

Course record

Sustainability Masterclass

3

Core files

Slides, Learner Guide and Lesson Plan

4

Activities

One folder per activity with scenario, questions and data

5

Calculator

alfredang.github.io/sgcarboncalculator

6

Reference

Keep the Learner Guide and source register to hand

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SCHEDULE

Learning Journey — Two Days

DAY 1

- Measure What Matters — Boundary, Scopes and Hotspots
  - Standards, boundaries and scopes
  - Inventory calculations and data quality
  - Hotspot analysis and economics
  - Activities 1–3

DAY 2

- Move From Insight to Action — Priorities, Plans and Governance
  - Scenarios, technologies and portfolio choices
  - EMS, governance and Singapore context
  - Action planning and capstone
  - Activities 4–9

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THREE CONNECTED TOPICS

Course Map

1

Topic 1 · Analyse Carbon Emission Hotspots

Standards, boundaries, data, scopes and hotspot diagnosis

2

Topic 2 · Prioritise Reduction Initiatives

Projection, technology, economics and transparent decisions

3

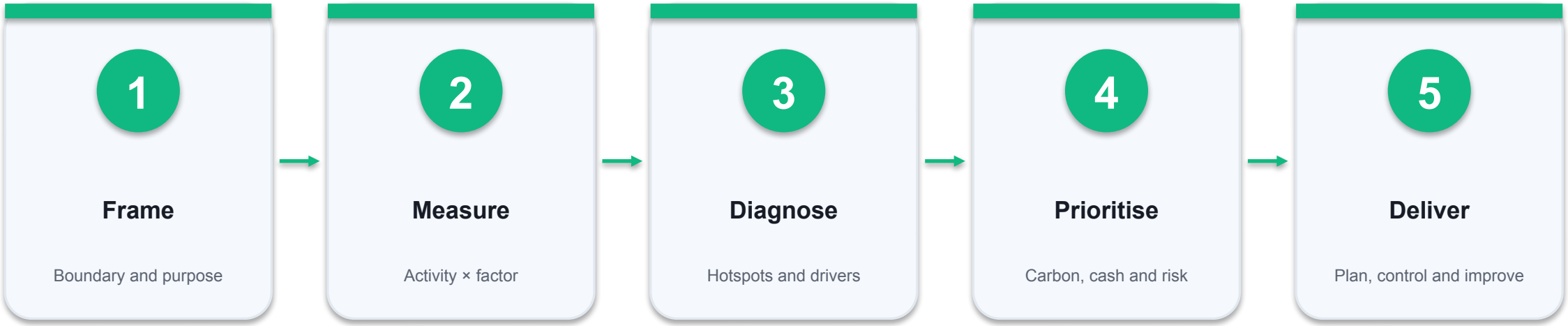
Topic 3 · Develop the Reduction Plan

EMS governance, action planning, Singapore requirements and review

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COURSE SPINE

The Carbon Management Decision Cycle



CORE IDEA

Every activity strengthens one part of the same auditable decision trail.

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START WITH PURPOSE

Carbon accounting is not the finish line.

It is the evidence system that helps people choose and deliver better reductions.

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1

TOPIC 1

Analyse Carbon Emission Hotspots

Standards, boundaries, data, scopes and hotspot diagnosis

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STANDARDS

A Carbon Inventory Is a Decision System

1

Purpose

Translate physical activity into comparable tCO_2e for decisions.

2

Boundary

State which entities, operations, gases and period are included.

3

Method

Use consistent activity data, emission factors and global-warming potentials.

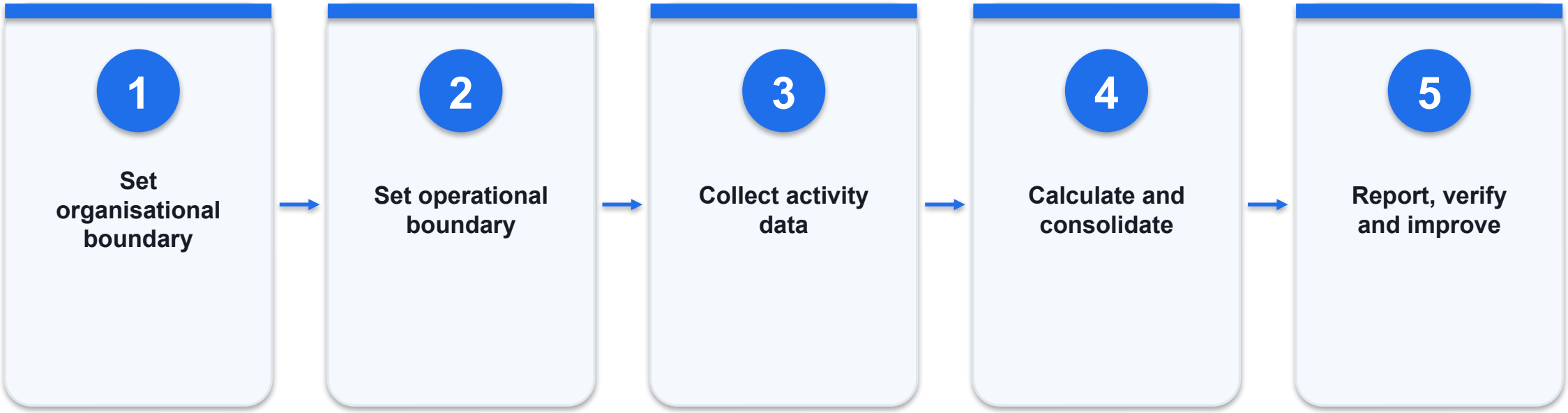
4

Evidence

Retain sources, assumptions, controls and recalculation rules.

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Build a Corporate Inventory in Five Moves



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PRINCIPLES

Five Principles Protect Decision Quality

1 **Relevance**
Reflect the organisation and users' needs.

2 **Completeness**
Account for all sources within the boundary; disclose exclusions.

3 **Consistency**
Enable meaningful comparison over time.

4 **Transparency**
Disclose methods, assumptions and evidence.

5 **Accuracy**
Reduce bias and uncertainty as far as practical.

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Organisational Boundary Choices

Equity share

- Account according to economic interest
- Useful where ownership shares matter
- Apply consistently across investments

Control approach

- Account for operations under financial or operational control
- Often aligns with management systems
- Document the chosen control test

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How the Standards Fit Together



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ASSURANCE

Accounting Standard vs Management Standard

GHG Protocol / ISO 14064

- What and how much is emitted
- Boundaries, categories, calculations
- Inventory quality and reporting

ISO 14001 EMS

- How environmental performance is managed
- Policy, objectives, controls and review
- Continual improvement through PDCA

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GASES

The Seven Greenhouse Gases

1. **CO_2**
   Fuel and process emissions
2. **CH_4**
   Waste, agriculture and fugitive emissions
3. **N_2O**
   Combustion, fertiliser and processes
4. **HFCs**
   Refrigeration and cooling
5. **PFCs**
   Electronics and industrial processes
6. **SF_6**
   Electrical equipment
7. **NF_3**
   Electronics and semiconductor manufacturing

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CORE CALCULATION

From Activity to tCO_2e

A transparent inventory keeps every assumption visible.

```
Emissions (tCO₂e) = Activity data ×
Emission factor × GWP ÷ unit conversion
```

CHECK 1

Example: kWh × kg CO_2e/kWh ÷ 1,000

CHECK 2

Use the factor for the correct geography and year

CHECK 3

Record factor source, unit and uncertainty

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DATA QUALITY

A Practical Data-Quality Ladder

1

1 · Supplier-specific

Verified primary data for the exact product or service.

2

2 · Metered activity

Invoices, meters, fuel records or logistics data.

3

3 · Engineering estimate

Measured drivers with documented assumptions.

4

4 · Spend proxy

Currency × environmentally extended input-output factor.

5

5 · Unsupported guess

Use only temporarily; flag for improvement.

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SCOPES

Three Scopes Prevent Category Confusion

1

Scope 1

Direct emissions from sources owned or controlled.

2

Scope 2

Indirect emissions from purchased electricity, steam, heat or cooling.

3

Scope 3

Other value-chain emissions upstream and downstream.

4

Rule

Scopes describe responsibility for accounting—not moral blame.

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SCOPE 2

Location-Based and Market-Based Views

Location-based

- Average grid emissions where consumption occurs
- Useful for physical system context
- Singapore example: EMA grid factor

Market-based

- Contractual instruments and supplier attributes
- Requires quality criteria and evidence
- Report both where applicable

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SCOPE 3

Follow the Value Chain, Not the Org Chart

1

Upstream

Purchased goods, capital goods, fuels, transport, waste, travel, commuting and leased assets.

2

Downstream

Distribution, processing, use, end-of-life, leased assets, franchises and investments.

3

Screen first

Estimate all relevant categories before investing in precision.

4

Engage next

Improve supplier and customer data at material hotspots.

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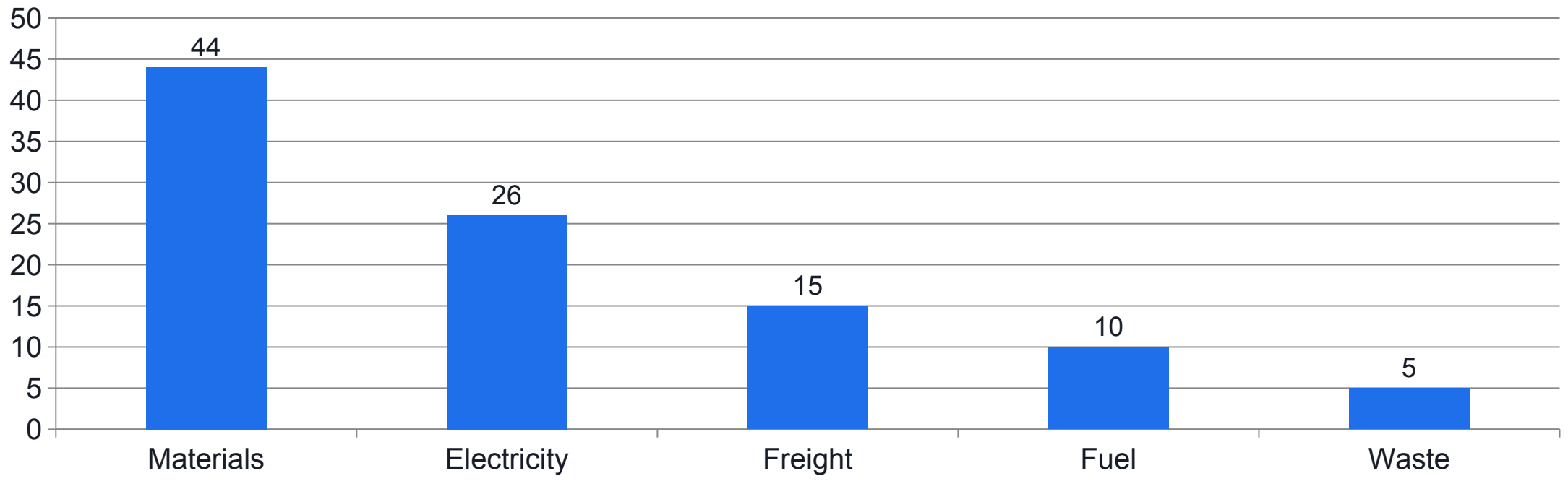
The Hotspot Analysis Funnel



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PARETO

Illustrative Footprint by Source



WHAT THE DATA SHOWS

Top two sources represent 70% of the screening footprint; validate them first.

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MATERIALITY

A Hotspot Is More Than the Largest Bar

1

Magnitude

Absolute or intensity-based share of emissions.

2

Influence

Ability to change technology, demand or supplier behaviour.

3

Risk

Carbon price, regulation, disruption or reputation.

4

Strategic fit

Connection to customers, growth and operational priorities.

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CONTROLS

Common Inventory Failure Modes

1

Missing sources

Completeness check against sites, ledgers and processes.

2

Double counting

Clear ownership, category definitions and consolidation rules.

3

Unit errors

Automated validation and unit conversion controls.

4

Wrong factor

Versioned factor library with geography and year.

5

Silent estimates

Estimate register with owner and replacement date.

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CALCULATOR FIT

Can the SG Carbon Calculator Be Used Here?



YES / FIT

Personal engagement: use calculator outputs to start conversations and identify behaviours.

NO / NOT YET

Corporate inventory: use controlled organisational activity data, documented factors and boundary rules.

The calculator is a screening and learning aid—not a substitute for a GHG Protocol corporate inventory.

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SINGAPORE

Reading Singapore Electricity Data Correctly

1

Published factor

EMA reported 0.402 kg CO_2/kWh for 2024.

2

Correct year

Match the factor year to the reporting period.

3

Correct unit

Convert kg to tonnes only once.

4

Record evidence

Save the factor source and access date.

5

Recalculate

Define when base-year electricity emissions are restated.

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ECONOMICS

Separate Carbon Value From Cash Value

1

Carbon effect

Annual and lifetime tCO_2e avoided.

2

Cash effect

Capex, opex, savings and residual value.

3

Risk effect

Carbon tax, energy volatility and compliance exposure.

4

Strategic effect

Resilience, customer value and learning.

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PAYBACK

Simple Payback Is Fast—but Incomplete

Useful as a screening metric when cash flows are stable.

```
Simple payback (years) = Initial investment
÷ Annual net cash saving
```

CHECK 1

Ignores time value of money

CHECK 2

Ignores cash flows after payback

CHECK 3

Never rank long-lived projects by payback alone

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NPV

NPV Makes Timing Explicit

Discount future net cash flows to present value.

$$NPV = \Sigma \left[\text{Net cash flow}_t \div (1+r)^t \right] - \text{Initial investment}$$

CHECK 1

Positive NPV creates financial value at the chosen discount rate

CHECK 2

Test energy price, utilisation and maintenance assumptions

CHECK 3

Use sensitivity analysis instead of false precision

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ABATEMENT COST

Compare Projects per Tonne Avoided

Marginal abatement cost exposes the price of each carbon outcome.

```
MAC = Lifetime net cost ÷ Lifetime tCO₂e avoided
```

CHECK 1

Negative cost means the measure saves money

CHECK 2

Use consistent lifetime and discounting

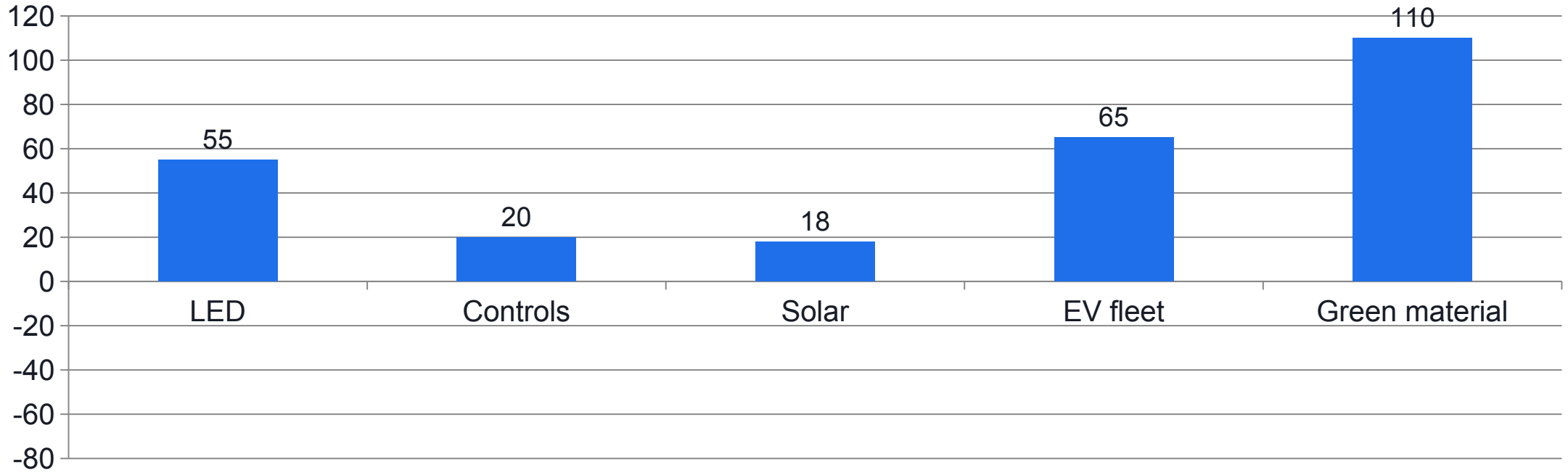
CHECK 3

Do not ignore feasibility and dependencies

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MACC

Illustrative Abatement Portfolio



WHAT THE DATA SHOWS

Negative-cost measures can help fund strategic measures with positive abatement cost.

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SENSITIVITY

Stress-Test the Variables That Matter

1. **Energy price**
   Low, expected and high tariff cases.

2. **Utilisation**
   Operating hours, throughput and occupancy.

3. **Performance**
   Degradation, rebound and implementation shortfall.

4. **Carbon value**
   Tax, internal carbon price or customer premium.

5. **Discount rate**
   Financing and risk assumptions.

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Two Ways to Price Carbon Internally

Shadow price

- Apply a hypothetical carbon value in investment cases
- Changes decisions without charging business units
- Good for stress-testing long-lived assets

Internal fee

- Charge emissions and recycle funds into reductions
- Creates a visible operating signal
- Needs governance and acceptance

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WORKED EXAMPLE

LED Retrofit Decision

Illustrative teaching data: S$120,000 capex; S$38,000 annual saving; 72 tCO_2e avoided per year; 8-year life.

```
Payback = 120,000 ÷ 38,000 = 3.16 years
Simple lifetime MAC = (120,000 − 8×38,000)
÷ (8×72) = −S$319/tCO₂e
```

CHECK 1

The project saves money and carbon

CHECK 2

Confirm maintenance, hours and baseline

CHECK 3

Use discounted cash flow for approval

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From Standards to a Defensible Hotspot

1

Frame

Purpose, organisational boundary, operational boundary and period.

2

Quantify

Activity data × factor × GWP with controlled units.

3

Diagnose

Rank magnitude, trend, risk and controllability.

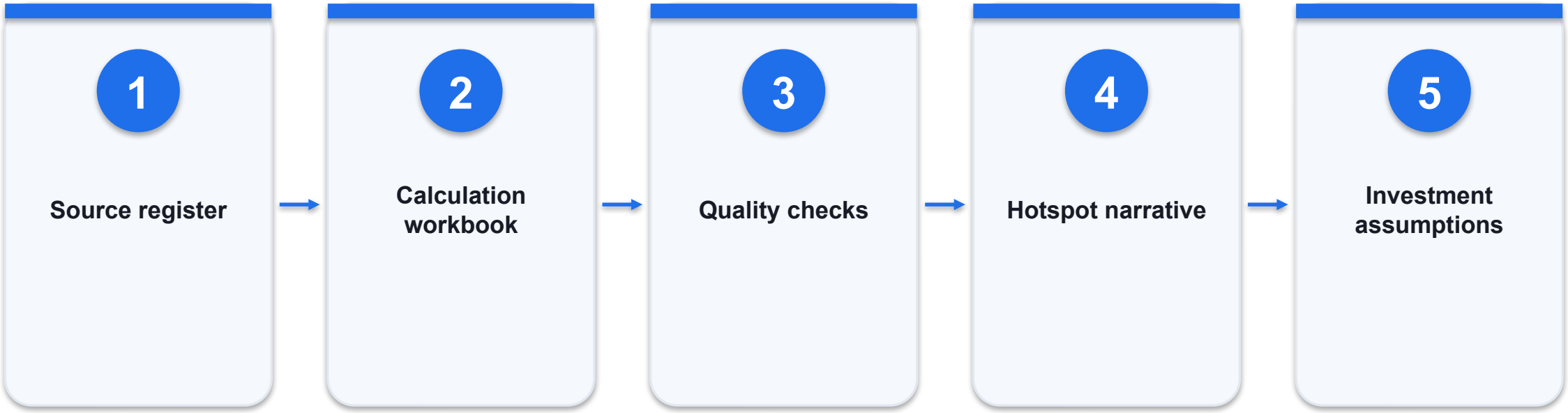
4

Value

Combine tonnes, cash, risk and strategic fit.

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Evidence Should Survive the Handover



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TOPIC 1 · TAKEAWAY

Precision without completeness can produce a confidently wrong hotspot.

Screen broadly, then improve the data where decisions are material.

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REAL-USE-CASE ACTIVITY

Map a Multi-Site Corporate Inventory

ACTIVITY 1 Define organisational and operational boundaries, classify sources and identify data gaps.

Define organisational and operational boundaries, classify sources and identify data gaps.

YOU'LL BUILD

Boundary memo, scope map and data-gap register

TOOLCHAIN

Case: Singapore retail and distribution group Source lens: GHG Protocol Corporate Standard

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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REAL-USE-CASE ACTIVITY

Use the SG Carbon Calculator as an Engagement Tool

ACTIVITY 2 Use personal-footprint screening responsibly, then translate behavioural insights into corporate data needs.

Use personal-footprint screening responsibly, then translate behavioural insights into corporate data needs.

YOU'LL BUILD

Calculator evidence, limitations note and engagement experiment

TOOLCHAIN

Case: Employee carbon-literacy campaign Source lens: SG Carbon Calculator and UN ActNow

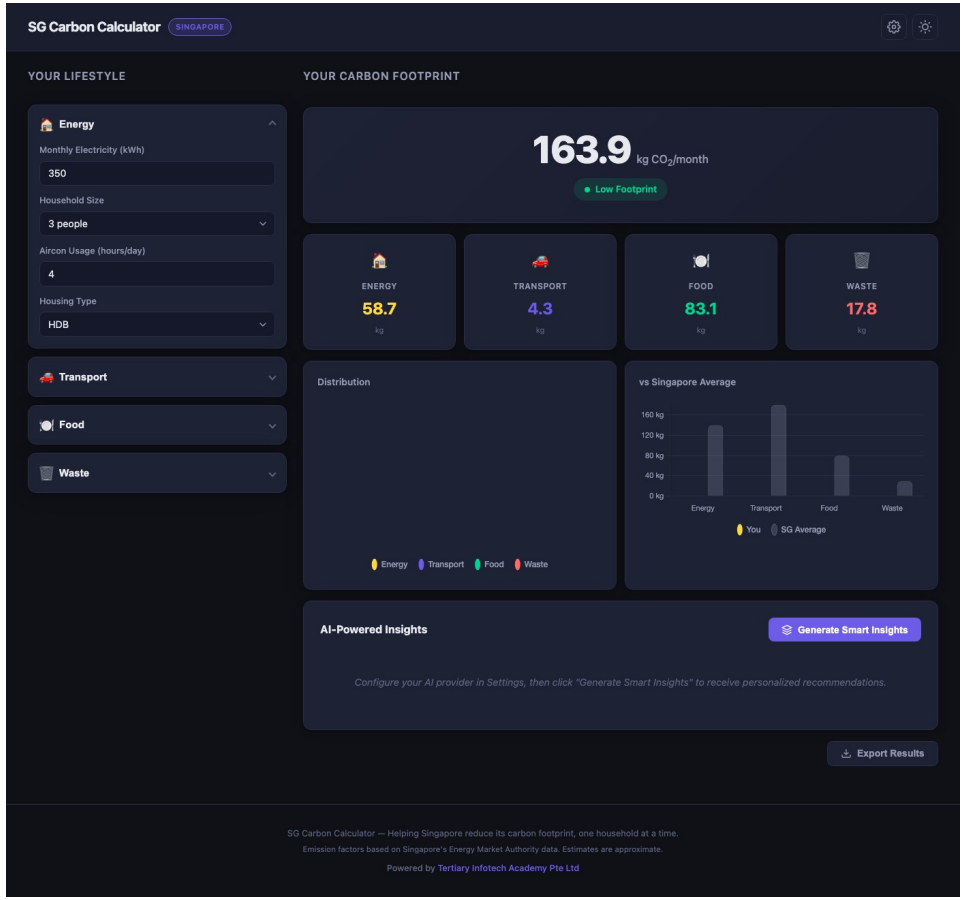
DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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ACTIVITY 2 · AUTHENTIC TOOL

SG Carbon Calculator: Use It for Screening, Not Inventory



RIGHT TOOL · RIGHT PURPOSE

- Explore how energy, transport, food and waste choices affect a personal screening result.
- Use fictional personas; avoid collecting unnecessary personal information.
- Translate insights into engagement hypotheses and operational data needs.
- Do not aggregate personal results into a corporate GHG inventory.

alfredang.github.io/sgcarboncalculator

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REAL-USE-CASE ACTIVITY

Find Hotspots in a Farm-to-Table Value Chain

ACTIVITY 3 Calculate a screening footprint, rank hotspots and propose a data-improvement plan.

Calculate a screening footprint, rank hotspots and propose a data-improvement plan.

YOU'LL BUILD

Calculated footprint, Pareto chart and hotspot recommendation

TOOLCHAIN

Case: CP Foods-inspired teaching case Source lens: SAP customer spotlight and GHG Protocol

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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LUNCH BREAK

30 minutes

2

TOPIC 2

Prioritise Reduction Initiatives

Projection, technology, economics and transparent decisions

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BASELINE

A Baseline Is a Managed Reference Point

1

Base year

Representative, reliable and relevant to the target.

2

Structural change

Acquisition, divestment or outsourcing may require recalculation.

3

Method change

Material factor or methodology changes may require restatement.

4

Growth

Separate absolute emissions from intensity and activity growth.

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PROJECTION

Start With the Business-as-Usual Case

Make growth, efficiency and factor assumptions explicit.

```
Future emissions = Future activity × Future emission factor × (1 − efficiency improvement)
```

CHECK 1

Build the no-action case first

CHECK 2

Avoid claiming reductions caused by grid decarbonisation as operational…

CHECK 3

Use compatible units and periods

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TARGETS

Absolute and Intensity Targets Answer Different Questions

Absolute target

- Total tCO_2e falls against baseline
- Directly aligned to atmospheric outcome
- Can challenge a fast-growing business

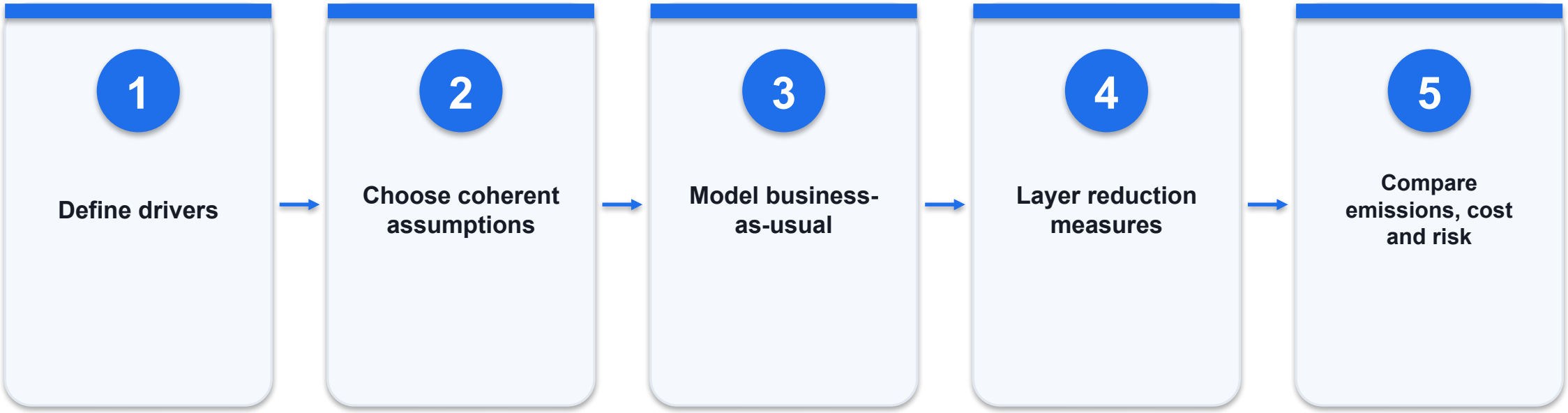
Intensity target

- tCO_2e per unit of output or revenue
- Separates growth from efficiency
- Can improve while total emissions rise

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SCENARIOS

Build Scenarios as Decision Stories



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UNCERTAINTY

Use Ranges Where the Future Is Uncertain

1

Low case

Slower growth, lower adoption or conservative performance.

2

Expected case

Best current estimate with documented evidence.

3

High case

Faster growth, price shock or implementation upside.

4

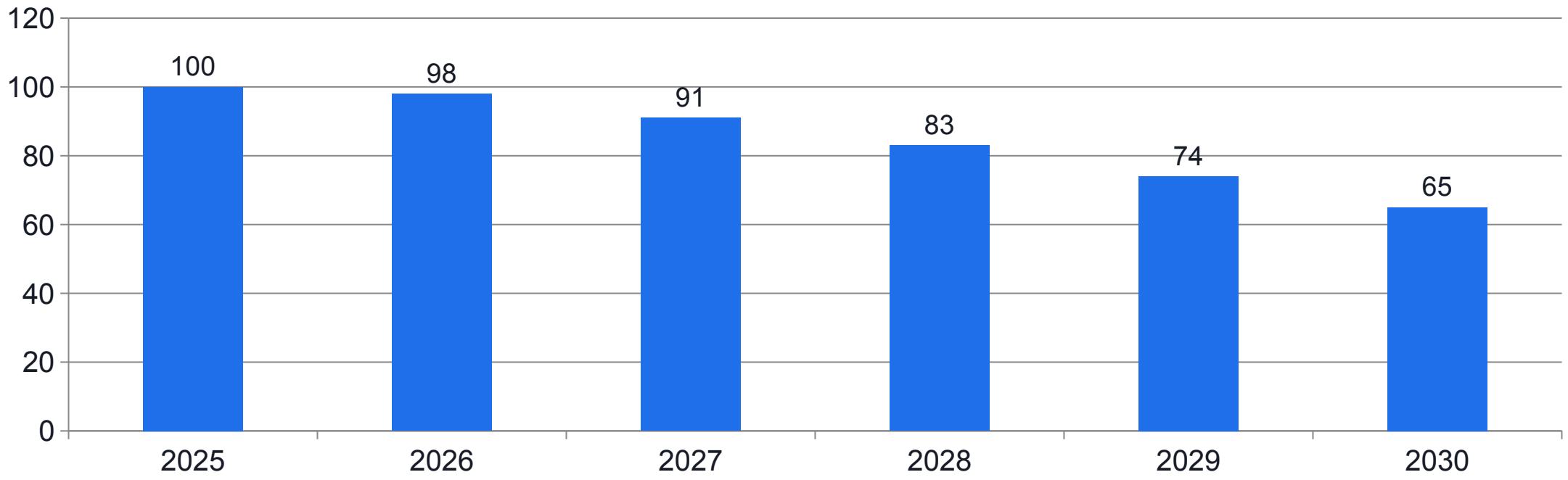
Trigger

Define the observation that causes a plan revision.

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TRAJECTORY

Illustrative 2030 Emissions Path



WHAT THE DATA SHOWS

A target path becomes actionable only when initiatives, owners and milestones explain each step down.

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DECOMPOSITION

Explain Why Emissions Changed

Separate operational performance from structural and market effects.

```
Δ Emissions = Activity effect + Efficiency
effect + Factor effect + Structural effect
```

CHECK 1

Activity effect captures volume or growth

CHECK 2

Efficiency effect captures operational improvement

CHECK 3

Factor effect captures energy or supplier decarbonisation

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REBOUND

Avoid the Rebound Trap

1

Efficiency gain

Less energy per service unit.

2

Behaviour response

Lower cost can increase demand or operating hours.

3

Net result

Gross saving minus additional consumption.

4

Control

Track both intensity and absolute activity.

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WORKED EXAMPLE

Electrifying a Singapore Delivery Fleet

Illustrative: 20 vans × 35,000 km/year; diesel van 0.25 kg CO_2e/km; EV 0.20 kWh/km; grid factor 0.402 kg CO_2/kWh.

```
Diesel = 175 tCO₂e/year
EV electricity = 56.3 tCO₂e/year
Screening reduction = 118.7 tCO₂e/year
```

CHECK 1

Add vehicle and battery lifecycle where material

CHECK 2

Model future grid factors separately

CHECK 3

Validate actual route, charging and payload data

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REDUCTION HIERARCHY

Reduce Before You Compensate

1

Avoid

Remove unnecessary demand, travel, material or process.

2

Reduce

Improve efficiency and yield.

3

Substitute

Use lower-carbon energy, material or transport.

4

Engage

Work with suppliers, customers and employees.

5

Neutralise residuals

Use credible removals only after deep reductions.

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ENERGY

Energy Levers: Measure, Control, Replace

1

Measure

Sub-meter material loads and normalise for production.

2

Control

Schedules, set-points, maintenance and automation.

3

Upgrade

Efficient motors, chillers, compressed air and heat recovery.

4

Replace

Renewable electricity and electrified heat where feasible.

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OPERATIONS

Operational Excellence Is Carbon Management

1. **Yield**
   Prevent scrap and rework.
2. **Flow**
   Reduce waiting, idle loads and unnecessary movement.
3. **Maintenance**
   Keep assets at designed efficiency.
4. **Planning**
   Match batches, routes and capacity to demand.
5. **Design**
   Remove carbon before it is locked into the process.

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VALUE CHAIN

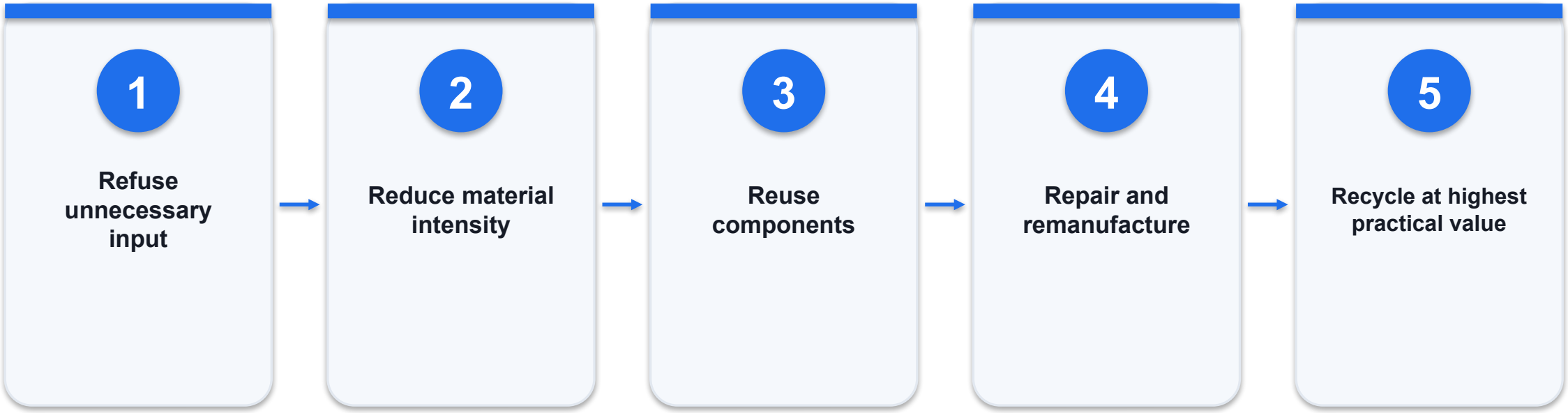
Supplier Engagement Needs a Data Ladder

1. **Screen**
   Use secondary factors to find material categories.
2. **Request**
   Ask material suppliers for product or process data.
3. **Validate**
   Check boundary, method, period and assurance.
4. **Collaborate**
   Co-design lower-carbon specifications and logistics.
5. **Contract**
   Embed evidence and performance expectations.

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CIRCULARITY

Circular Design Changes the Carbon Equation



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DIGITAL

Industry 4.0 Can Help—and Create New Risks

Opportunity

- IoT enables real-time energy and process data
- Analytics detects anomalies and predicts failures
- ERP connects carbon to materials, orders and suppliers

Implementation risk

- High integration cost and interoperability gaps
- Data security and ownership concerns
- Automation can scale poor-quality data

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ERP

What an Enterprise Footprint Platform Connects

1

Transactions

Materials, purchase orders, production and logistics.

2

Factors

Versioned libraries matched to geography and period.

3

Flows

Energy and material movement across products and sites.

4

Results

Corporate and product footprints with drill-down.

5

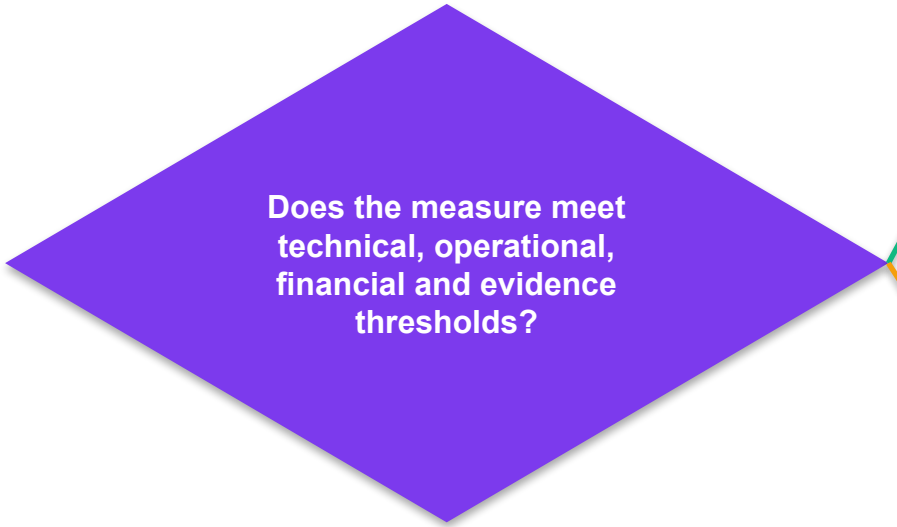
Action

Hotspots integrated into operational decisions.

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FEASIBILITY

Is the Technology Ready for This Site?



YES / FIT

Pilot with success criteria, monitoring plan and scale-up gate.

NO / NOT YET

Redesign, sequence a dependency, or retain in the opportunity backlog.

A promising technology is not yet a feasible initiative until site constraints and implementation risk are tested.

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PILOT

Design a Pilot That Can Be Judged

1. **Baseline**
   Comparable pre-intervention period and operating conditions.

2. **Intervention**
   Specific asset, control or behaviour change.

3. **Metric**
   Energy, output, emissions, cost and quality guardrail.

4. **Duration**
   Long enough to capture representative variation.

5. **Decision rule**
   Scale, modify or stop threshold agreed in advance.

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PRIORITISATION

A Portfolio Needs More Than One Score

1

Impact

Annual and cumulative tCO_2e.

2

Economics

NPV, payback and abatement cost.

3

Feasibility

Technology maturity, resources and disruption.

4

Strategic fit

Customer, compliance and resilience value.

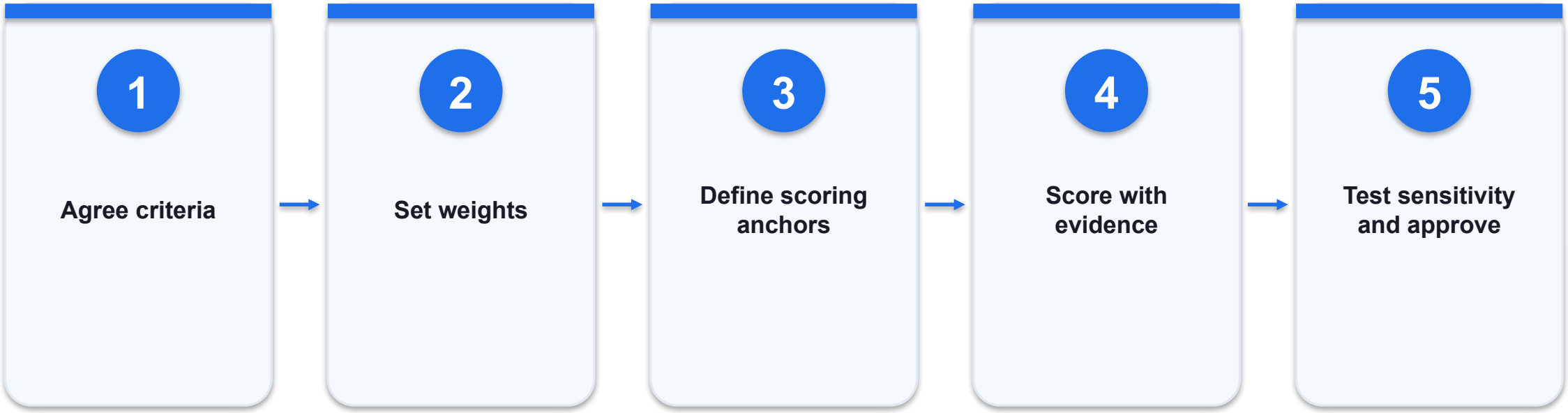
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Confidence

Data quality and dependency risk.

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A Transparent Multi-Criteria Decision Process



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WEIGHTED SCORE

Make Trade-Offs Visible

Use a common scale and documented evidence.

```
Weighted score = Σ (criterion score × criterion weight)
```

CHECK 1

Weights must total 100%

CHECK 2

Use behavioural anchors for 1–5 scores

CHECK 3

Do not let the formula replace governance

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PORTFOLIO

No-Regret Measures and Strategic Bets

No-regret

- Strong economics and proven technology
- Low dependency and quick implementation
- Build momentum and fund later work

Strategic bet

- Higher uncertainty or capital need
- Enables deep reductions or future compliance
- Use staged gates and learning milestones

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SEQUENCING

What Should Start First?



YES / FIT

Place in the immediate delivery wave with a named owner and budget gate.

NO / NOT YET

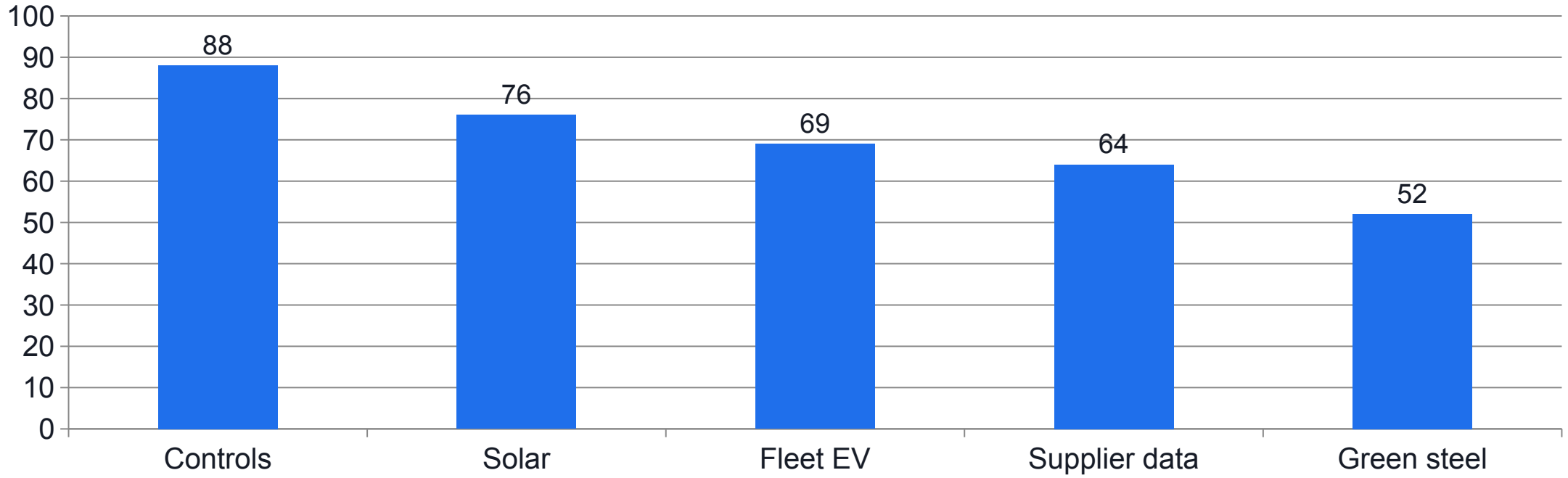
Place in enable, pilot, partner or monitor wave with a trigger date.

Sequence dependencies: data platform before optimisation, charging before fleet conversion, specification before supplier tender.

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PORTFOLIO VIEW

Illustrative Initiative Scores



WHAT THE DATA SHOWS

A score supports discussion; the approval record must still explain constraints, dependencies and dissent.

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GOVERNANCE

Counter Common Decision Biases

1. **Payback bias**
   Include lifecycle value and residual risk.
2. **Pet-project bias**
   Require comparable evidence and scoring anchors.
3. **Precision bias**
   Show ranges and confidence ratings.
4. **Silo bias**
   Include operations, finance, procurement and users.
5. **Offset bias**
   Apply the reduction hierarchy first.

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WORKED EXAMPLE

Portfolio Choice Under a S$500k Cap

Illustrative measures: controls S$80k/160 t; solar S$280k/210 t; EV pilot S$140k/90 t; green material S$220k/300 t.

```
Possible Wave 1: controls + solar + EV
pilot = S$500k and 460 tCO₂e/year screened
benefit
```

CHECK 1

Green material has high impact but needs procurement validation

CHECK 2

Check double counting and dependencies

CHECK 3

Retain a reserve for measurement and commissioning

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TOPIC 2 · RECAP

Prioritisation Is a Portfolio Conversation

1. **Project**
   Model baseline, intervention and uncertainty.

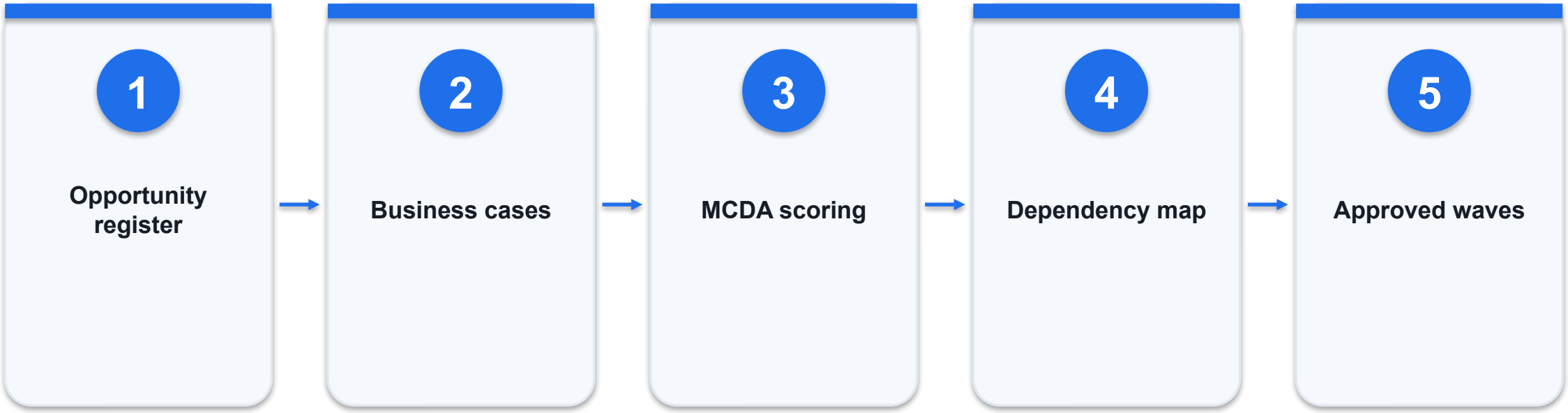
2. **Prove**
   Test technical and operating feasibility.

3. **Compare**
   Use carbon, cash, risk, strategy and confidence.

4. **Sequence**
   Balance no-regret delivery with deep-reduction enablers.

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From Long List to Funded Roadmap



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TOPIC 2 · TAKEAWAY

The best portfolio is credible enough to fund and flexible enough to learn.

Use scenarios, pilots and stage gates to turn uncertainty into managed evidence.

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REAL-USE-CASE ACTIVITY

Model a Singapore Fleet Transition

ACTIVITY 4 Build business-as-usual and reduction scenarios for diesel, route optimisation and fleet electrification.

Build business-as-usual and reduction scenarios for diesel, route optimisation and fleet electrification.

YOU'LL BUILD

2030 scenario model, assumptions register and recommendation

TOOLCHAIN

Case: LionCity Logistics Source lens: EMA grid statistics and GHG Protocol

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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REAL-USE-CASE ACTIVITY

Screen a Smart-Factory Carbon Initiative

ACTIVITY 5 Assess the technical, operational, data and financial feasibility of IoT/ERP-enabled energy optimisation.

Assess the technical, operational, data and financial feasibility of IoT/ERP-enabled energy optimisation.

YOU'LL BUILD

Feasibility scorecard and pilot charter

TOOLCHAIN

Case: Matsumoto Precision-inspired teaching case Source lens: SAP customer spotlight and Yurtay (2025)

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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REAL-USE-CASE ACTIVITY

Prioritise a Reduction Portfolio

ACTIVITY 6 Combine MACC, weighted criteria, dependencies and budget constraints into a transparent portfolio.

Combine MACC, weighted criteria, dependencies and budget constraints into a transparent portfolio.

YOU'LL BUILD

MACC, MCDA matrix, funded roadmap and decision record

TOOLCHAIN

Case: WITTE Automotive-inspired teaching case Source lens: SAP customer spotlight, CarbonChain and IMD

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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3

TOPIC 3

Develop the Reduction Plan

EMS governance, action planning, Singapore requirements and review

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An EMS Turns Intent Into Repeatable Control

1. **Context**
   Interested parties, risks, opportunities and obligations.

2. **Leadership**
   Policy, accountability and resources.

3. **Planning**
   Aspects, objectives, targets and actions.

4. **Support & operation**
   Competence, communication and operational controls.

5. **Evaluation & improvement**
   Monitor, audit, review and correct.

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Carbon Management Through PDCA



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GOVERNANCE

Ownership and Accountability Are Different

Process owner

- Controls the activity and operational data
- Implements agreed actions
- Explains performance variance

Carbon / GHG manager

- Maintains method, inventory and controls
- Coordinates reporting and assurance
- Escalates quality and compliance issues

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CONTROLS

Build Controls Around the Data Journey

1. **Capture**
   Authorised sources and complete populations.

2. **Transform**
   Locked factors, units and calculation rules.

3. **Review**
   Threshold checks and independent approval.

4. **Change**
   Version control and recalculation log.

5. **Retain**
   Evidence repository and retention period.

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COMPETENCE

The Plan Needs People Who Can Operate It

1. **Executives**
   Approve ambition, capital and risk appetite.

2. **Finance**
   Validate business cases and reporting controls.

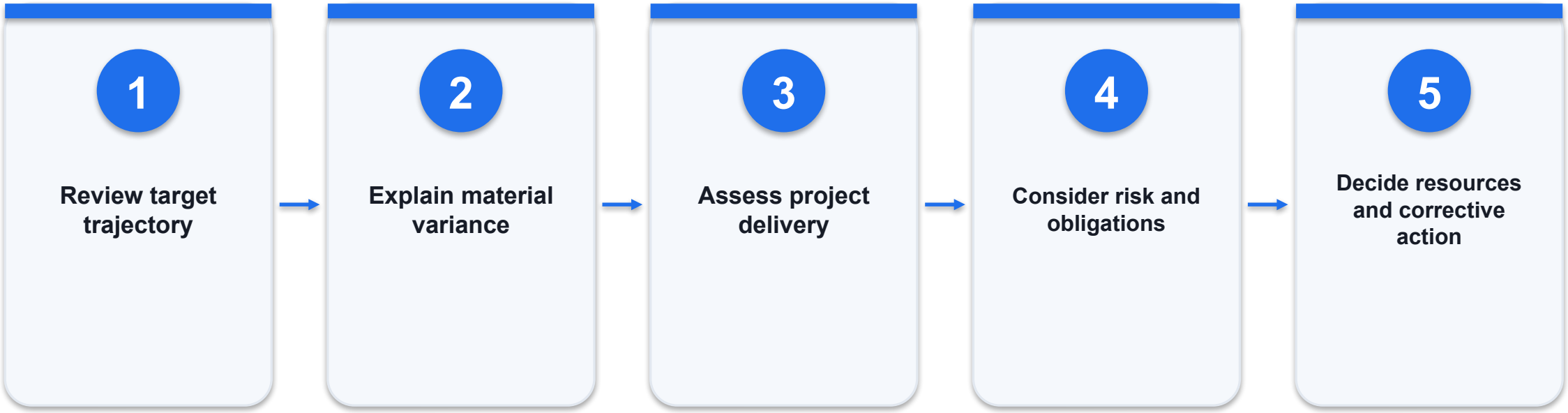
3. **Operations**
   Own sources, projects and root causes.

4. **Procurement**
   Engage suppliers and embed specifications.

5. **Data / IT**
   Maintain architecture, access and quality.

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A Useful Review Drives Decisions



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AUDIT

Audit the System, Not Only the Number

1. **Boundary**
   Is it complete and consistently applied?

2. **Evidence**
   Can samples be traced to source?

3. **Calculation**
   Are factors, units and formulas controlled?

4. **Action**
   Are corrective actions owned and closed?

5. **Communication**
   Are claims supported and balanced?

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CORRECTIVE ACTION

Is the Variance a Data Error or Performance Gap?



YES / FIT

Fix the data/control issue, assess restatement and prevent recurrence.

NO / NOT YET

Investigate operational root cause and implement a performance corrective action.

Do not hide a missed target by changing the data; do not launch a project to fix a calculation error.

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TAKEAWAY

A footprint becomes manageable when it has owners, controls and review rhythms.

The inventory is an operating process—not an annual spreadsheet event.

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PLAN ARCHITECTURE

A Credible Reduction Plan Has Eight Parts

1

1 · Boundary and baseline

What is covered and measured.

2

2 · Hotspots

Where emissions concentrate and why.

3

3 · Target

Outcome, year, scope and trajectory.

4

4 · Initiatives

Measures, savings and dependencies.

5

5 · Roadmap

Sequence, milestones and resources.

6

6 · Governance

Owners and decision rights.

7

7 · Measurement

KPIs, evidence and review.

8

8 · Risk & communication

Uncertainty, claims and escalation.

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TARGET DESIGN

A Target Needs Scope and Substance

Weak target

- Reduce emissions significantly
- No base year or boundary
- No trajectory or owner

Decision-ready target

- 35% absolute Scope 1+2 reduction by 2030 from 2025
- Boundary and recalculation policy stated
- Annual trajectory and accountable executive

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ROADMAP

Sequence the Reduction Plan



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KPI TREE

Connect Outcome, Driver and Delivery Metrics

1

Outcome

Total tCO_2e and tCO_2e per functional unit.

2

Driver

kWh/unit, fuel/km, kg material/unit, waste yield.

3

Delivery

Project milestone, installed capacity, supplier coverage.

4

Quality

Primary-data share, estimate share and control exceptions.

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RISK

Make Failure Modes Visible Before Launch

1

Delivery

Delay, skills, vendor or permit.

2

Performance

Savings lower than modelled.

3

Commercial

Tariff, price, demand or finance.

4

Data

Missing evidence or incompatible systems.

5

Claims

Overstatement, double counting or unclear offsets.

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CLAIMS

Reduction and Offset Are Not Interchangeable

Operational reduction

- Changes emissions within the inventory or value chain
- Traceable to a measure and baseline
- Supports the reduction trajectory

Offset / credit

- Finances a reduction or removal outside the inventory
- Requires quality and retirement evidence
- Disclose separately from gross emissions

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COMMUNICATION

A Responsible Claim Shows the Full Picture

1. **Boundary**
   Which entities, scopes and period?

2. **Metric**
   Gross, net, absolute or intensity?

3. **Method**
   Which standard, factors and recalculations?

4. **Progress**
   Actual result versus trajectory.

5. **Limitations**
   Estimates, uncertainty and exclusions.

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TARGET PATH

Translate a 35% Target Into Annual Milestones

Illustrative 2025 baseline: 10,000 tCO_2e. 2030 target: 6,500 tCO_2e.

```
Required reduction = 3,500 tCO₂e
Linear planning guide = 700 tCO₂e/year
average
```

CHECK 1

Actual roadmap need not be linear

CHECK 2

Link each step change to initiatives

CHECK 3

Track business-as-usual growth separately

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RESOURCING

Budget the Enablers, Not Only the Hardware

1. **Measurement**
   Meters, factors, assurance and data engineering.

2. **People**
   Project management, training and supplier engagement.

3. **Change**
   Communications, workflow redesign and incentives.

4. **Contingency**
   Performance shortfall and dependency risk.

5. **Maintenance**
   Commissioning and sustained optimisation.

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ACTION PLAN

Every Action Needs a Complete Contract

1. **What**
   Specific intervention and boundary.
2. **Why**
   Hotspot and expected outcome.
3. **Who**
   Accountable owner and support roles.
4. **When**
   Start, milestone, completion and review.
5. **How much**
   Budget, resources and quantified benefit.
6. **Evidence**
   KPI, source and verification method.

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SMARTER

Make Objectives SMARTER

A target should guide action and adaptation.

```
Specific · Measurable · Achievable ·
Relevant · Time-bound · Evaluated ·
Reviewed
```

CHECK 1

Use an outcome metric and at least one driver metric

CHECK 2

Name the evidence source and review cadence

CHECK 3

Predefine escalation and revision conditions

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Clarify Roles Before Delivery Starts

1. Accountable owner approves
2. Responsible team delivers
3. Consulted experts shape decisions
4. Informed stakeholders receive updates

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REGISTER

Use a Single Initiative Register

1. **Identity**
   Initiative ID, source, site and owner.

2. **Value**
   Capex, savings, tCO_2e and confidence.

3. **Delivery**
   Stage, milestone, dependency and issue.

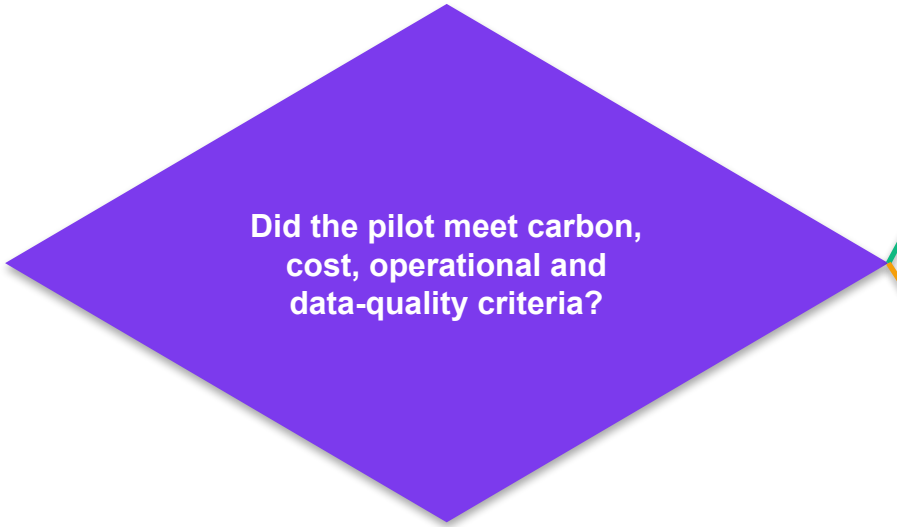
4. **Decision**
   Approval, gate and rationale.

5. **Evidence**
   Baseline, M&V method and realised result.

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STAGE GATE

Should the Initiative Scale?



YES / FIT

Approve rollout with updated benefit case and implementation controls.

NO / NOT YET

Modify or stop; document learning and update the opportunity register.

Stage gates protect capital while preserving learning from uncertainty.

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M&V

Measure Realised Savings Fairly

1. **Baseline model**
   Representative pre-project performance.
2. **Adjustment**
   Production, weather, occupancy or route mix.
3. **Boundary**
   Avoid leakage and double counting.
4. **Persistence**
   Check that savings continue after launch.
5. **Approval**
   Independent review of material results.

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SINGAPORE CARBON TAX

Know the Current Singapore Thresholds

1

Reportable facility

Annual direct emissions at or above 2,000 tCO_2e.

2

Taxable facility

Annual direct emissions at or above 25,000 tCO_2e.

3

2026–2027 tax rate

S$45 per tCO_2e.

4

Toward 2030

Government has indicated S$50–S$80 per tCO_2e by 2030.

5

Scope

Industrial facilities in specified sectors; verify current CPA requirements.

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Reportable and Taxable Facilities

≥2,000 tCO_2e direct

- Register as reportable where applicable
- Submit annual Emissions Report by 30 June following year
- Appoint required representatives and manager

≥25,000 tCO_2e direct

- Also subject to carbon tax
- Monitoring Plan and verified Emissions Report
- Stronger controls and verification requirements

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CARBON TAX

Screen the 2026 Carbon-Tax Exposure

Illustrative taxable emissions: 30,000 tCO_2e. 2026 rate: S$45/$tCO_2e$.

```
Gross screened liability = 30,000 × S$45 =
S$1,350,000
```

CHECK 1

Use verified reportable emissions and current law for filing

CHECK 2

Account for eligible allowances only with competent advice

CHECK 3

Use the result to stress-test investments—not as the inventory boundary

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A Plan Is a Managed System

1

Govern

Owners, controls, competence and management review.

2

Plan

Boundary, target, initiatives, roadmap and resources.

3

Deliver

Action contracts, stage gates and dependencies.

4

Learn

Measurement, audit, corrective action and transparent claims.

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From Target to Continual Improvement



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TOPIC 3 · TAKEAWAY

The reduction plan is credible when someone can execute it, finance it and verify it.

Targets become performance only through governance and evidence.

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REAL-USE-CASE ACTIVITY

Prepare a Singapore MR and Carbon-Tax Readiness Plan

ACTIVITY 7 Screen regulatory status, data controls, reporting obligations and carbon-price exposure.

Screen regulatory status, data controls, reporting obligations and carbon-price exposure.

YOU'LL BUILD

Obligations matrix, control plan and screened tax exposure

TOOLCHAIN

Case: Jurong specialty-chemicals facility Source lens: NEA Carbon Tax and MR requirements

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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REAL-USE-CASE ACTIVITY

Design an Employee ActNow Programme

ACTIVITY 8 Convert selected personal actions into a privacy-respecting corporate engagement programme with measurable outcomes.

Convert selected personal actions into a privacy-respecting corporate engagement programme with measurable outcomes.

YOU'LL BUILD

Programme canvas, measurement plan and communication copy

TOOLCHAIN

Case: Singapore professional-services firm Source lens: United Nations ActNow

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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REAL-USE-CASE ACTIVITY

Build the Board-Ready Carbon Reduction Plan

ACTIVITY 9 Integrate hotspot analysis, prioritisation, governance and action planning into a defensible 2030 plan.

Integrate hotspot analysis, prioritisation, governance and action planning into a defensible 2030 plan.

YOU'LL BUILD

Ten-slide board pack plus action register and management-review dashboard

TOOLCHAIN

Case: Lion City Precision Pte Ltd
Source lens: Integrated course case using GHG Protocol, ISO/EMS and Singapore context

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

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EVIDENCE BASE

Standards, Regulation and Further Reading

Verify current requirements before external reporting

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REFERENCES · 1–4 · FULL LINKS IN LEARNER GUIDE

Source Register

1. **GHG Protocol Corporate Standard**
   ghgprotocol.org

2. **United Nations ActNow — Ten Actions**
   un.org

3. **NEA — Singapore Carbon Tax**
   nea.gov.sg

4. **NEA — Measurement and Reporting Requirements**
   nea.gov.sg

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REFERENCES · 5–8 · FULL LINKS IN LEARNER GUIDE

Source Register

1. **EMA — Singapore Energy Statistics, Chapter 2**
   ema.gov.sg

2. **SAP Sustainability Footprint Management**
   sap.com

3. **IMD — Carbon Management**
   imd.org

4. **Yurtay (2025), Industry 4.0 and ERP Systems**
   mdpi.com

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Source Register

1. **Synesgy — Carbon Footprint Guide**
   synesgy.com

2. **Green Earth — Carbon Footprints**
   green.earth

3. **CarbonChain — Reducing Company Footprints**
   carbonchain.com

4. **SG Carbon Calculator**
   github.com

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WRAP-UP

From Footprint to Managed Reduction

Integrate measurement, decisions and delivery

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LEARNING OUTCOMES

What You Can Now Do

1

LO1

Analyse emission hotspots with defensible boundaries and data.

2

LO2

Prioritise a portfolio using scenarios, feasibility, economics and transparent criteria.

3

LO3

Develop an executable reduction plan with targets, governance and measurement.

4

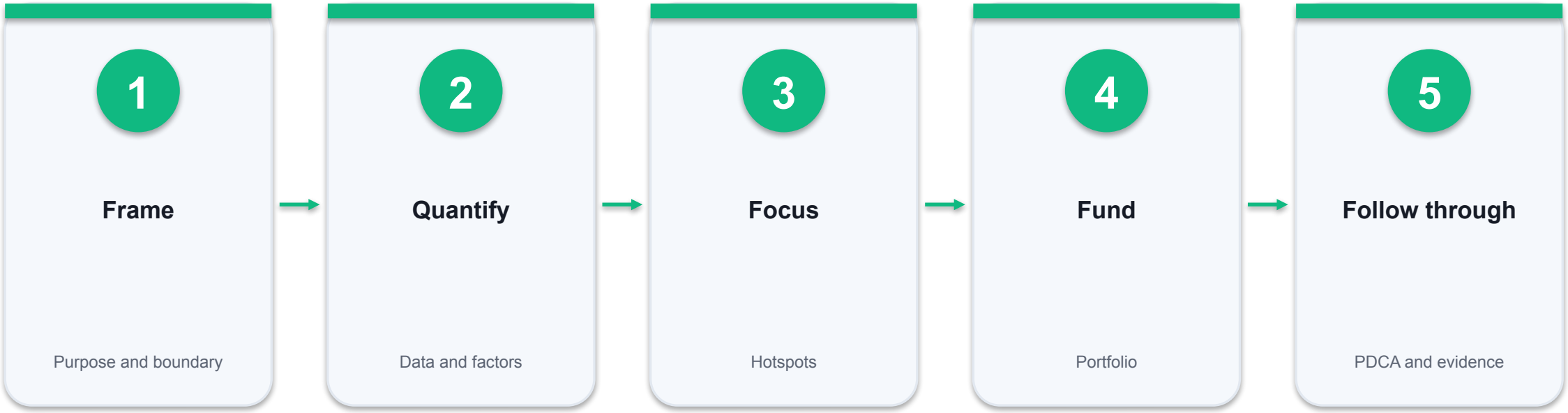
Professional judgement

State assumptions, uncertainty and limitations without hiding them.

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TAKE IT BACK TO WORK

Your Practical Playbook



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CONTINUE LEARNING

Recommended Courses

1. Sustainability Reporting and Assurance
2. Environmental, Social and Governance (ESG) Essentials
3. Data Analytics for Sustainability
4. Business Process Automation with Power Automate
5. Business Innovation with Artificial Intelligence

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COURSE FEEDBACK

Course Feedback & Certificate

1

Feedback

Complete the course feedback survey before you leave.

2

Improve

Tell us what worked and what to improve.

3

Support

Ask the trainer if you cannot open the survey or files.

4

Certificate

Your certificate of completion will be emailed to you.

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SUSTAINABILITY MASTERCLASS

Thank You

Measure what matters. Reduce what matters most. Prove what changed.

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